

Schematic V1.1.0

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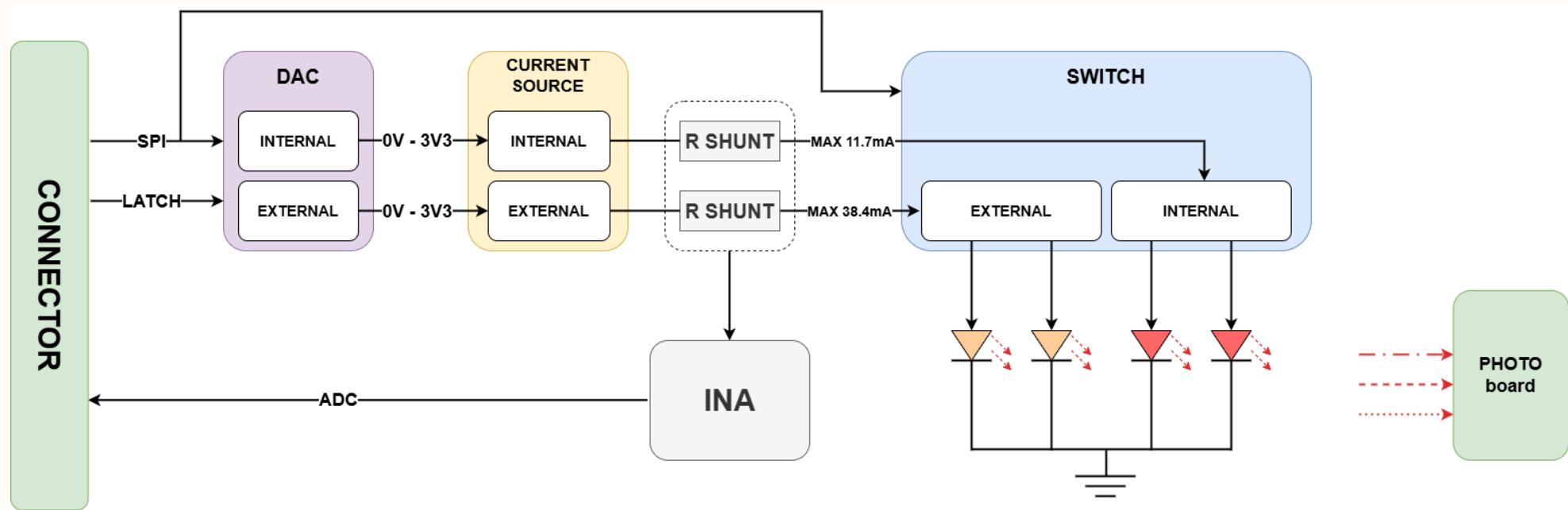
PAGE	COMPONENT/ FUNCTION
001	INDEX
002	BLOCK_DIAGRAM
003	POWER SWITCH AND LDO
004	COMMUNICATION
005	CURRENT_SOURCE
006	SWITCHING

History

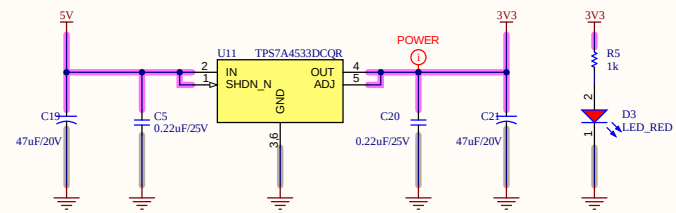
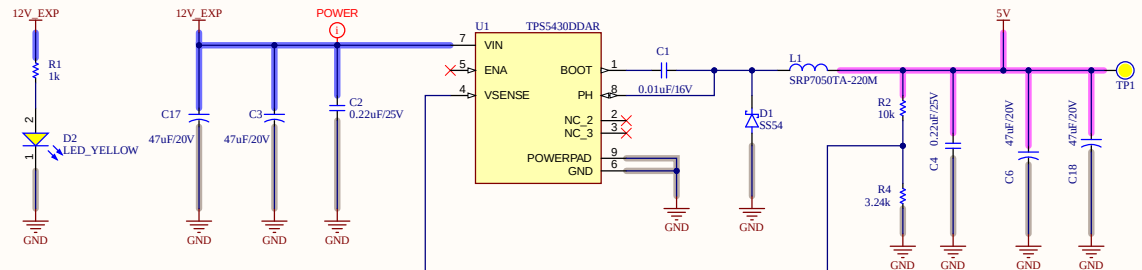
VERSION	DISCRIPTION
1.0.0	INITIAL PROJECT: LASER INT & EXT ON A BOARD
1.1.0	AJUST RSHUNT
1.2.0	CHANGE P FET TO AO3401, fix footprint internal laser



Project: BEE-PC2	Sub-system: LDA	
Design by: Duy Nguyen	Sheet title: 1_INDEX.SchDoc	Sheet 1 of 7
Checked by: Bao Bui	Document No: 1	Size: A3
Approved by: Vu Pham	Revision: 1.2.0	Date: 10-Aug-26

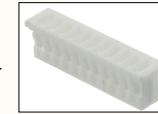


Project: BEE-PC2	Sub-system: LDA	
Design by: Duy Nguyen	Sheet title: 2_BLOCK_DIAGRAM.SchD	Sheet 2 of 7
Checked by: Bao Bui	Document No: 2	Size: A3
Approved by: Vu Pham	Revision: 1.2.0	Date: 10-Jul-26



Project: BEE-PC2	Sub-system: LDA	
Design by: Duy Nguyen	Sheet title: 3_POWER SWITCH AND LDO	Sheet 3 of 7
Checked by: Bao Bui	Document No: 3	Size: A3
Approved by: Vu Pham	Revision: 1.2.0	Date: 25-Jun-26

- ELECTRICAL:
 - CURRENT RATING: 1A/CONTACT
 - VOLTAGE RATING: 125V/CONTACT
- ENVIRONMENTAL:
 - TEMPERATURE RANGE: -40°C TO +105°C
- PHYSICAL:
 - PITCH-MATING: 0.049" (1.25mm)
- WIRE:
 - WIRE SIZE: 24 AWG
 - MAX CURRENT: 1A per Wire
 - ISULATION MATERIAL: PTFE
- MATING:
 - Molex - 0510211100



0510211100

EXT LASER SPI MOSI

EXT LASER SPI MISO

EXT LASER SPI SCK

EXT LASER DAC nCS

EXT LASER DAC LATCH

EXT LASER DAC nCS1

EXT LASER DAC

12V EXP

J2

1

2

3

4

5

6

7

8

9

10

11

D9 S2303

D34 S2303

D35 S2303

D44 S2303

GND

GND

GND

GND

GND

M1 M2

0532611171

- ELECTRICAL:
 - CURRENT RATING: 1A/CONTACT
 - VOLTAGE RATING: 125V/CONTACT
- ENVIRONMENTAL:
 - TEMPERATURE RANGE: -40°C TO +105°C
- PHYSICAL:
 - PITCH-MATING: 0.049" (1.25mm)
- WIRE:
 - WIRE SIZE: 24 AWG
 - MAX CURRENT: 1A per Wire
 - ISULATION MATERIAL: PTFE
- MATING:
 - Molex - 0510210800



0510210800

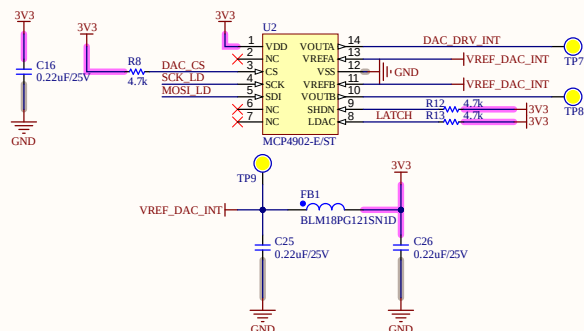
Pin	Function
01	MOSI_LD
02	EXT LASER SPI MOSI
03	MISO_LD
04	EXT LASER SPI MISO
05	DAC_CS
06	EXT LASER SPI SCK
07	SCK_LD
08	EXT LASER_DAC_nCS
09	DAC_CS
10	EXT LASER_DAC_LATCH
11	LATCH
12	EXT LASER_DAC_nCS
13	SW_INT_CS
14	EXT LASER_DAC_LATCH
15	SW_INT_CS
16	EXT LASER_DAC_nCS
17	SW_EXT_CS
18	EXT LASER_DAC_nCS
19	nMO_INT
20	EXT LASER_DAC
21	nMO_EXT
22	EXT LASER_DAC

CLP-111-02-X-D

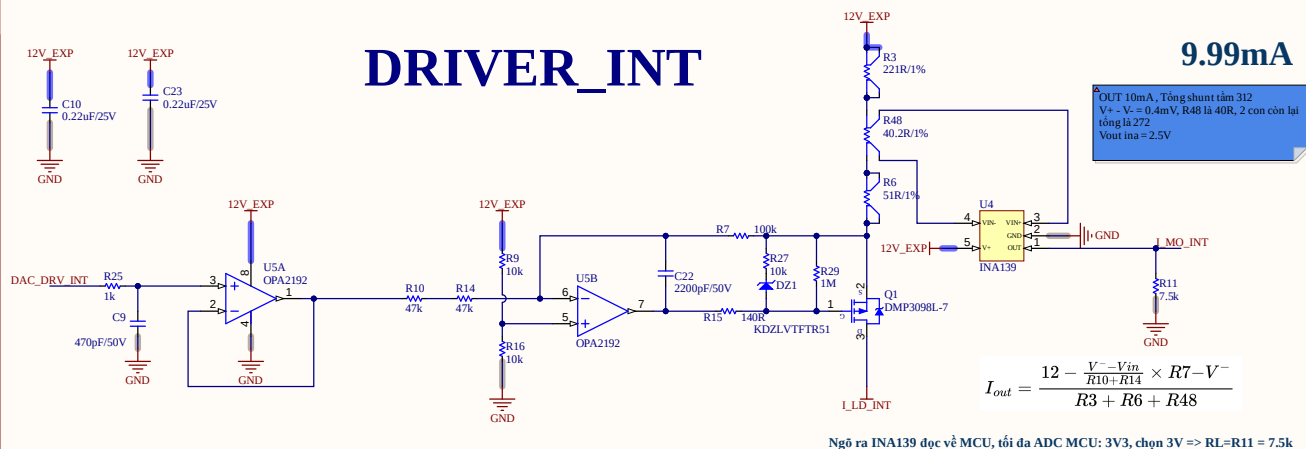


Date: 25-Jun-26

DAC



DRIVER_INT

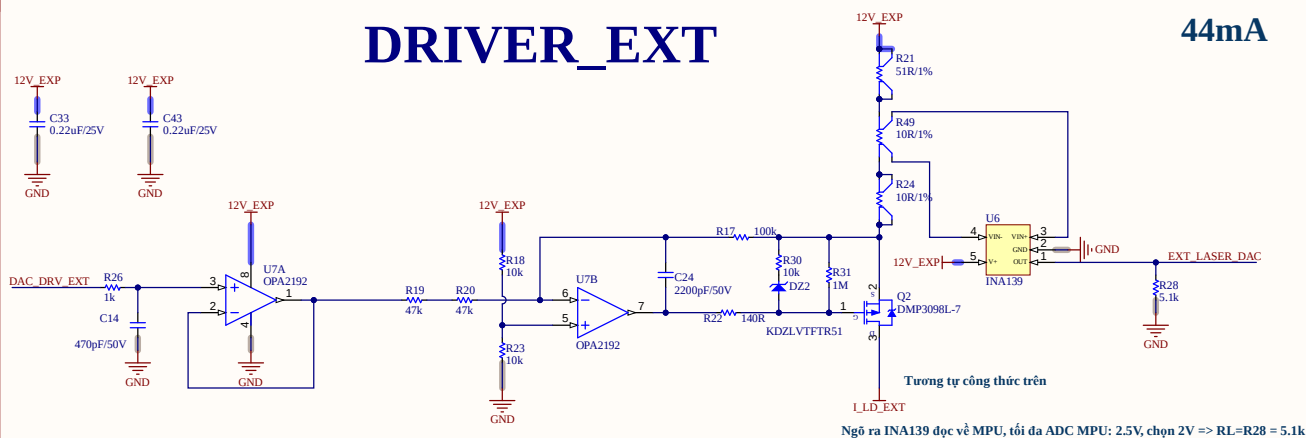


9.99mA

$$I_{out} = \frac{12 - \frac{V_{in}}{R_{10} + R_{14}} \times R_7 - V_{in}}{R_3 + R_6 + R_{48}}$$

Ngõ ra INA139 đọc về MCU, tối đa ADC MCU: 3V3, chọn 3V => RL=R11 = 7.5k

DRIVER_EXT



44mA

Tương tự công thức trên

Ngõ ra INA139 đọc về MPU, tối đa ADC MPU: 2.5V, chọn 2V => RL=R28 = 5.1k



Project: BEE-PC2

Sub-system: LDA

Design by: Duy Nguyen

Sheet title: 5_CURRENT_SOURCE.SchDoc Sheet 5 of 7

Checked by: Bao Bui

Document No: 5

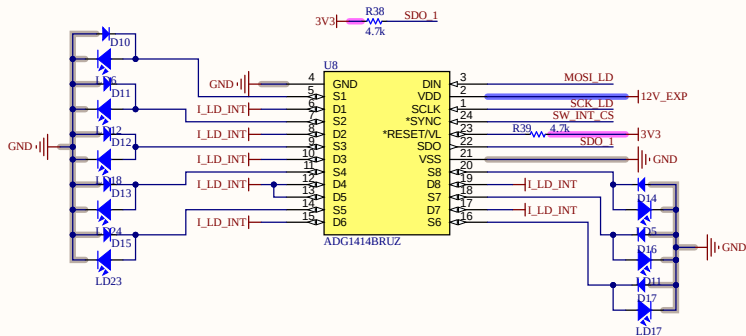
Size: A3

Approved by: Vu Pham

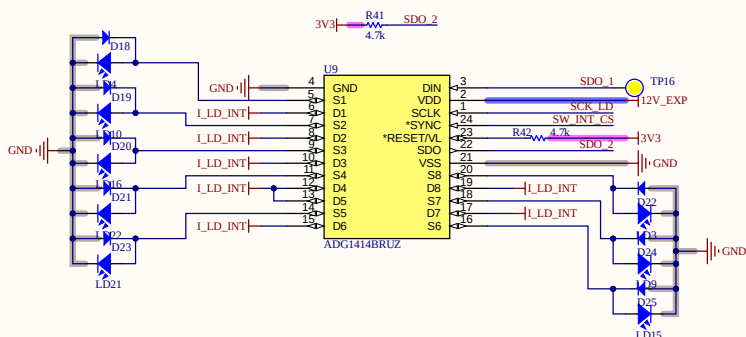
Revision: 1.2.0

Date: 11-Aug-26

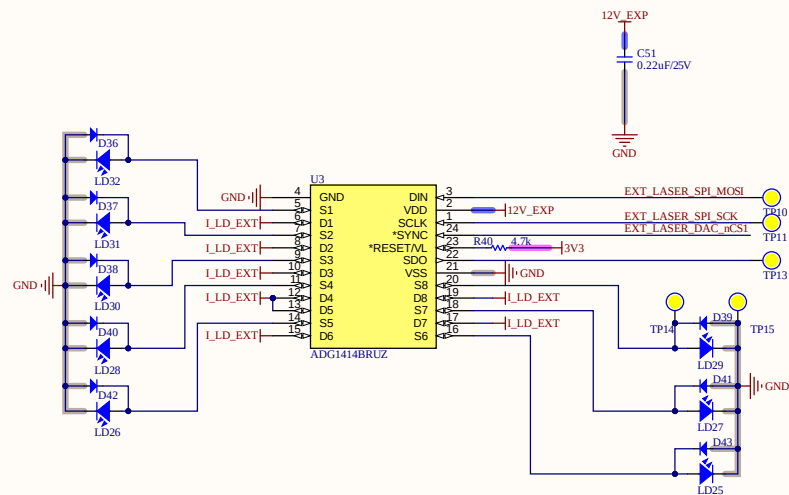
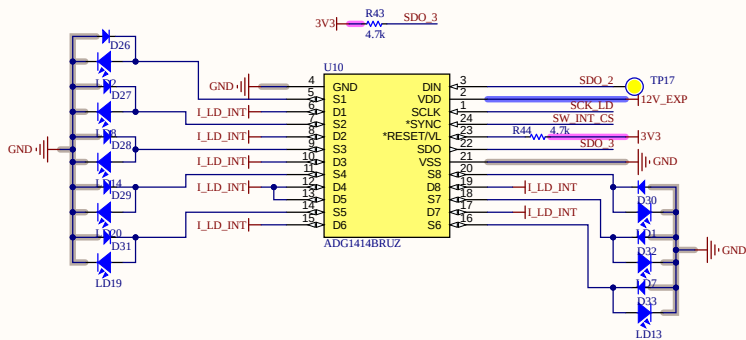
MOSI



SDO_1



SDO_2



Project: BEE-PC2

Sub-system: LDA

Design by: Duy Nguyen

Sheet title: 6_SWITCHING.SchDoc

Sheet 6 of 7

Checked by: Bao Bui

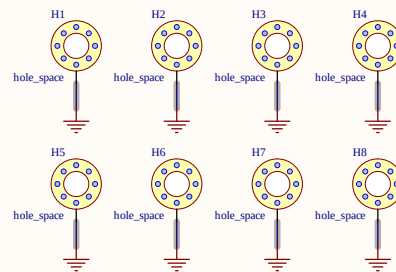
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Size: A3

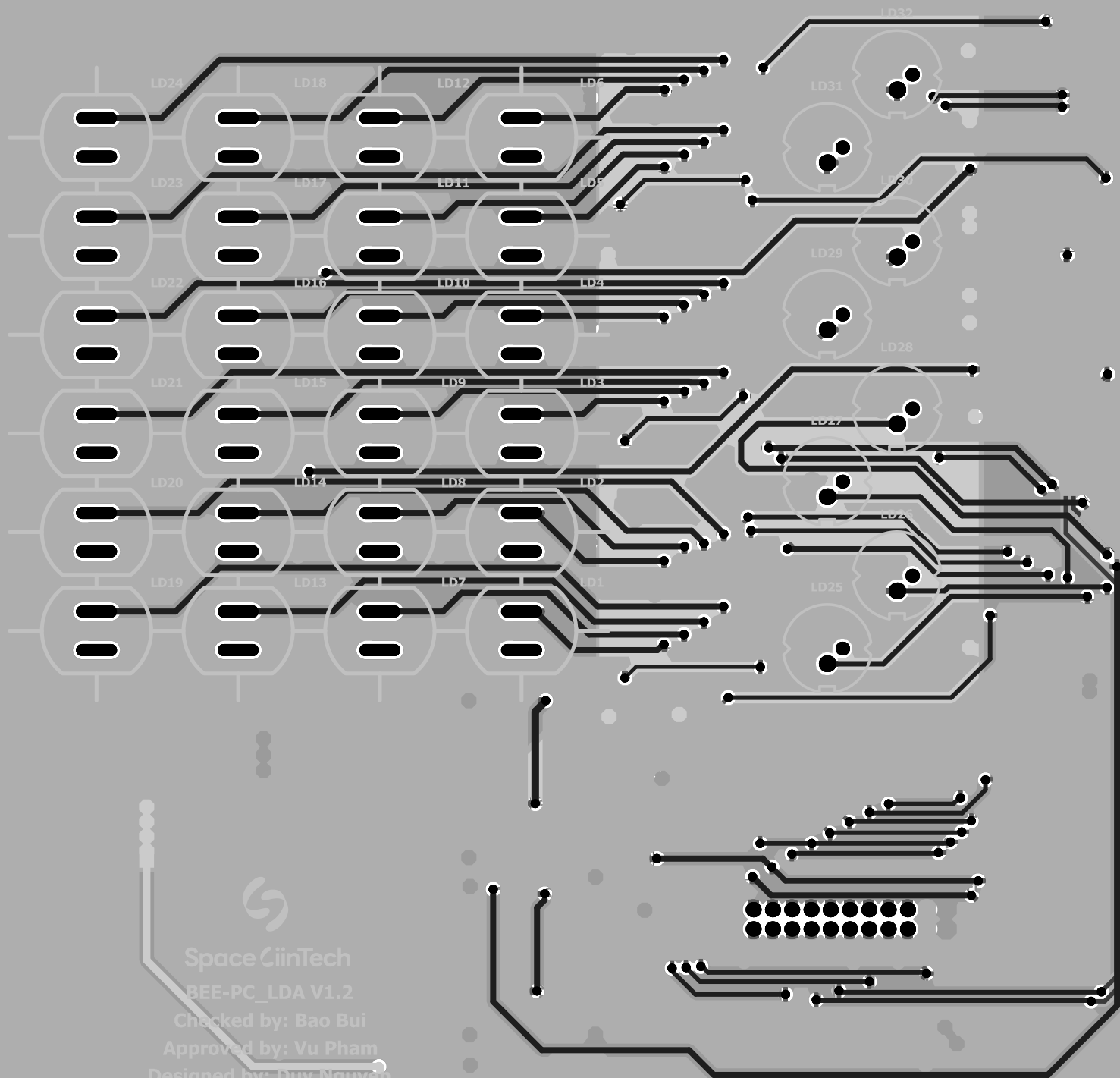
Approved by: Vu Pham

Revision: 1.2.0

Date: 25-Jun-26



Project: BEE-PC2		Sub-system: LDA	
Design by: Duy Nguyen	Sheet title: MOUNTING.SchDoc	Sheet 7 of 7	
Checked by: Bao Bui	Document No: 7	Size: A3	
Approved by: Vu Pham	Revision: 1.2.0	Date: 25-Jun-26	




Space CiiTech
BEE-PC_LDA V1.2
Checked by: Bao Bui
Approved by: Vu Pham
Designed by: Duy Nguyen

